

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only

REDTIBIN (DECITABINE) POWDER FOR INJECTION 50MG/VIAL

COMPOSITION

Each Single-dose Vial contains Decitabine 50 mg (Lyophilized)

Excipients:

Monobasic Potassium Phosphate, Sodium Hydroxide, Water for Injection, Nitrogen(Process Aid)

Product Description:

Decitabine Powder for Injection 50 mg/vial is a white to almost white lyophilized powder for reconstitution. The product is packed in 20 mL USP type I clear tubular glass vial, stoppered with 20 mm dark grey bromobutyl rubber stopper and sealed with 20 mm red colour aluminium flip-off seal.

Description of reconstituted solution: Clear colorless solution free from visible extraneous matter

CLINICAL INFORMATION

Therapeutic indications

Decitabine is indicated for treatment of patients with myelodysplastic syndromes (MDS) including previously treated and untreated, de novo and secondary MDS of all French-American-British (FAB) subtypes (refractory anemia, refractory anemia with ringed sideroblasts, refractory anemia with excess blasts, refractory anemia with excess blasts in transformation, and chronic myelomonocytic leukemia) and Intermediate-1, Intermediate-2, and High-Risk International Prognostic Scoring System (IPSS) groups.

For the treatment of adult patients aged 65 years and above with newly diagnosed de novo or secondary acute myeloid leukaemia (AML), according to the World Health Organisation (WHO) classification, who are not candidates for standard induction chemotherapy.

Posology and method of administration

Treatment of patients with myelodysplastic syndromes

There are two regimens for decitabine administration. With either regimen it is recommended that patients be treated for a minimum of 4 cycles; however, a complete or partial response may take longer than 4 cycles.

Complete blood counts and platelet counts should be performed as needed to monitor response and toxicity, but at a minimum, prior to each cycle. Liver chemistries and serum creatinine should be obtained prior to initiation of treatment.

Treatment Regimen - Option 1

Decitabine is administered at a dose of 15 mg/m² by continuous intravenous infusion over 3 hours repeated every 8 hours for 3 days. This cycle should be repeated every 6 weeks. Patients may be pre-medicated with standard anti-emetic therapy.

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If hematologic recovery ($ANC \geq 1,000/\mu L$ and platelets $\geq 50,000/\mu L$) from a previous decitabine treatment cycle requires more than 6 weeks, then the next cycle of decitabine therapy should be delayed and dosing temporarily reduced by following this algorithm:

- Recovery requiring more than 6, but less than 8 weeks – Decitabine dosing to be delayed for up to 2 weeks and the dose temporarily reduced to 11 mg/m^2 every 8 hours ($33 \text{ mg/m}^2/\text{day}$, $99 \text{ mg/m}^2/\text{cycle}$) upon restarting therapy.
- Recovery requiring more than 8, but less than 10 weeks – Patient should be assessed for disease progression (by bone marrow aspirates); in the absence of progression, the decitabine dose should be delayed up to 2 more weeks and the dose reduced to 11 mg/m^2 every 8 hours ($33 \text{ mg/m}^2/\text{day}$, $99 \text{ mg/m}^2/\text{cycle}$) upon restarting therapy, then maintained or increased in subsequent cycles as clinically indicated.

Treatment Regimen -Option 2

Decitabine is administered at a dose of 20 mg/m^2 by continuous intravenous infusion over 1 hour repeated daily for 5 days. This cycle should be repeated every 4 weeks. Patients may be pre-medicated with standard anti-emetic therapy.

If myelosuppression is present, subsequent treatment cycles of decitabine should be delayed until there is hematologic recovery ($ANC \geq 1,000/\mu L$ platelets $\geq 50,000/\mu L$).

Patients with Non-hematologic Toxicity

Following the first cycle of decitabine treatment, if any of the following non-hematologic toxicities are present, decitabine treatment should not be restarted until the toxicity is resolved: 1) serum creatinine $\geq 2 \text{ mg/dL}$; 2) SGPT, total bilirubin ≥ 2 times ULN; 3) and active or uncontrolled infection.

Treatment Regimen in Acute Myeloid Leukemia

In a treatment cycle, decitabine is administered at a dose of 20 mg/m^2 body surface area by intravenous infusion over 1 hour repeated daily for 5 consecutive days (i.e., a total of 5 doses per treatment cycle). The total daily dose must not exceed 20 mg/m^2 and the total dose per treatment cycle must not exceed 100 mg/m^2 . If a dose is missed, treatment should be resumed as soon as possible. The cycle should be repeated every 4 weeks depending on the patients clinical response and observed toxicity. It is recommended that patients be treated for a minimum of 4 cycles; however, a complete or partial remission may take longer than 4 cycles to be obtained. Treatment may be continued as long as the patient shows response, continues to benefit or exhibits stable disease, i.e., in the absence of overt progression.

If after 4 cycles, the patients haematological values (e.g., platelet counts or absolute neutrophil count), have not returned to pre-treatment levels or if disease progression occurs (peripheral blast counts are increasing or bone marrow blast counts are worsening), the patient may be considered to be a non-responder and alternative therapeutic options to decitabine should be considered.

Pre-medication for the prevention of nausea and vomiting is not routinely recommended but may be administered if required.

In AML Myelosuppression and adverse events related to myelosuppression (thrombocytopenia, anaemia, neutropenia, and febrile neutropenia) are common in both treated and untreated patients with AML. Complications of myelosuppression include infections and bleeding. Treatment may

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be delayed at the discretion of the treating physician, if the patient experiences myelosuppression-associated complications, such as those described below:

- Febrile neutropenia (temperature $\geq 38.5^{\circ}\text{C}$ and absolute neutrophil count $< 1,000/\mu\text{L}$)
- Active viral, bacterial or fungal infection (i.e., requiring intravenous anti-infectives or extensive supportive care)
- Hemorrhage (gastrointestinal, genito-urinary, pulmonary with platelets $< 25,000/\mu\text{L}$ or any central nervous system hemorrhage).

Treatment with decitabine may be resumed once these conditions have improved or have been stabilized with adequate treatment (anti-infective therapy, transfusions, or growth factors).

In clinical studies, approximately one-third of patients receiving decitabine required a dose-delay. Dose reduction is not recommended.

Special Populations

Pediatrics

The safety and effectiveness in pediatric patients have not been established.

Hepatic impairment

Studies in patients with hepatic impairment have not been conducted. The need for dosage adjustment in patients with hepatic impairment has not been evaluated. If worsening hepatic function occurs, patients should be carefully monitored.

Renal impairment

Studies in patients with renal impairment have not been conducted; however, data from clinical studies that included patients with mild-moderate impairment indicated no need for dosage adjustment. Patients with severe renal impairment were excluded from these studies.

Method of administration

Decitabine is administered by intravenous infusion. A central venous catheter is not required.

Special precautions for disposal and other handling

Recommendations for safe handling

Skin contact with the solution should be avoided and protective gloves must be worn. Standard procedures for dealing with anticancer agents should be adopted.

Reconstitution procedure

The powder should be aseptically reconstituted with 10 ml of water for injections. Upon reconstitution, each ml contains approximately 5 mg of decitabine at pH 6.7 to 7.3. Within 15 minutes of reconstitution, the solution must be further diluted with cold infusion fluids [sodium chloride 9 mg/ml (0.9%) solution for injection or 5% w/v Dextrose Injection and Lactated Ringer Injection] to a final concentration of 0.1 to 1.0 mg/ml.

Decitabine should not be infused through the same intravenous access/line with other medicinal products.

Disposal

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This medicinal product is for single use only. Any unused product or waste material should be disposed of in accordance with local requirements.

Contraindications

- Hypersensitivity to decitabine or to any of the excipients.
- Lactation.

Special warnings and precautions for use

Myelosuppression

Myelosuppression and complications of myelosuppression, including infections and bleeding that occur in patients with AML may be exacerbated with decitabine treatment. Therefore patients are at increased risk for severe infections (due to any pathogen such as bacterial, fungal and viral), with potentially fatal outcome. Patients should be monitored for signs and symptoms of infection and treated promptly.

In clinical studies, the majority of patients had baseline Grade 3/4 myelosuppression. In patients with baseline Grade 2 abnormalities, worsening of myelosuppression was seen in most patients and more frequently than in patients with baseline Grade 1 or 0 abnormalities. Myelosuppression caused by decitabine is reversible. Complete blood and platelet counts should be performed regularly, as clinically indicated and prior to each treatment cycle. In the presence of myelosuppression or its complications, treatment with decitabine may be interrupted and/or supportive measures instituted.

Treatment with decitabine is associated with neutropenia and thrombocytopenia. After administration of the recommended dosage for the first cycle, treatment for subsequent cycles should be adjusted. Clinicians should consider the need for early institution of growth factors and/or antimicrobial agents for the prevention or treatment of infections in patients with MDS. Myelosuppression and worsening neutropenia may occur more frequently in the first or second treatment cycles, and may not necessarily indicate progression of underlying MDS.

Respiratory, thoracic and mediastinal disorders

Cases of interstitial lung disease (ILD) (including pulmonary infiltrates, organising pneumonia and pulmonary fibrosis) without signs of infectious aetiology have been reported in patients receiving decitabine. Careful assessment of patients with an acute onset or unexplained worsening of pulmonary symptoms should be performed to exclude ILD. If ILD is confirmed, appropriate treatment should be initiated.

Hepatic impairment

The use of decitabine in patients with hepatic impairment has not been established. Caution should be exercised in the administration of decitabine to patients with hepatic impairment and patients should be monitored closely.

Renal impairment

The use of decitabine in patients with severe renal impairment has not been studied. Caution should be exercised in the administration of decitabine to patients with severe renal impairment (Creatinine Clearance [CrCl] < 30 ml/min) and these patients should be monitored closely.

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Cardiac disease

Patients with a history of severe congestive heart failure or clinically unstable cardiac disease were excluded from clinical studies and therefore the safety and efficacy of decitabine in these patients has not been established.

Effects on ability to drive and use machines

Decitabine may have moderate influence on the ability to drive and use machines. Patients should be advised that they may experience undesirable effects such as anaemia during treatment. Therefore, caution should be recommended when driving a car or operating machines.

Excipients

This medicine contains 0.5 mmol potassium per vial. After reconstitution and dilution of the solution for intravenous infusion, this medicine contains between 1-10 mmol potassium per dose depending on the infusion fluid for dilution. To be taken into consideration by patients with reduced kidney function or patients on a controlled potassium diet.

This medicine contains 0.29 mmol sodium per vial. After reconstitution and dilution of the solution for intravenous infusion, this medicine contains between 0.6-6 mmol sodium per dose depending on the infusion fluid for dilution. To be taken into consideration by patients on a controlled sodium diet.

Interaction with other medicinal products and other forms of interaction

No formal clinical drug interaction studies with decitabine have been conducted.

There is the potential for a drug-drug interaction with other agents which are also activated by sequential phosphorylation (via intracellular phosphokinase activities) and/or metabolized by enzymes implicated in the inactivation of decitabine (e.g., cytidine deaminase). Therefore, caution should be exercised if these drugs are combined with decitabine.

Impact of co-administered medicinal products on decitabine

Cytochrome (CYP) 450-mediated metabolic interactions are not anticipated as decitabine metabolism is not mediated by this system but by oxidative deamination.

Impact of decitabine on co-administered medicinal products

Given its low *in vitro* plasma protein binding (< 1%), decitabine is unlikely to displace co-administered medicinal products from their plasma protein binding. Decitabine has been shown to be a weak inhibitor of P-gp mediated transport *in vitro* and is therefore also not expected to affect P-gp mediated transport of co-administered medicinal products.

Fertility, pregnancy and lactation

Contraception in males and females

Women of childbearing potential must use effective contraceptive measures and avoid becoming pregnant while being treated with Decitabine. The time period following treatment with Decitabine where it is safe to become pregnant is unknown. Men should use effective contraceptive measures and be advised to not father a child while receiving Decitabine, and for 3 months following completion of treatment.

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The use of decitabine with hormonal contraceptives has not been studied.

Pregnancy

There are no adequate data on the use of Decitabine in pregnant women. Studies have shown that decitabine is teratogenic in rats and mice. The potential risk for humans is unknown. Based on results from animal studies and its mechanism of action, decitabine should not be used during pregnancy and in women of childbearing potential not using effective contraception. If decitabine is used during pregnancy, or if a patient becomes pregnant while receiving this medicinal product, the patient should be apprised of the potential hazard to the foetus.

Lactation

It is not known whether decitabine or its metabolites are excreted in breast milk. Decitabine is contraindicated during lactation; therefore if treatment with Decitabine is required, breast-feeding must be discontinued.

Fertility

No human data on the effect of decitabine on fertility are available. In non-clinical animal studies, decitabine alters male fertility and is mutagenic. Because of the possibility of infertility as a consequence of Decitabine therapy, men should seek advice on conservation of sperm and female patients of childbearing potential should seek consultation regarding oocyte cryopreservation prior to initiation of treatment with decitabine.

Undesirable effects

Myelodysplastic syndromes (MDS)

Most Commonly Occurring Adverse Reactions: neutropenia, thrombocytopenia, anemia, fatigue, pyrexia, nausea, cough, petechiae, constipation, diarrhea, and hyperglycemia.

Blood and lymphatic system disorders: Neutropenia, Thrombocytopenia, Anemia NOS, Febrile neutropenia, Leukopenia NOS, Lymphadenopathy, Thrombocythemia, Anemia, Leukopenia, Pancytopenia, myelosuppression, splenomegaly.

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Cardiac disorders: Pulmonary edema NOS, Cardiac failure congestive, Tachycardia, myocardial infarction, cardio-respiratory arrest, cardiomyopathy, atrial fibrillation, supraventricular tachycardia.

Eye disorders: Vision blurred

Ear and Labyrinth disorders: Ear pain

Gastrointestinal disorders: Nausea, Constipation, Diarrhea NOS, Vomiting NOS, Abdominal pain NOS, Oral mucosal petechiae, Stomatitis, Dyspepsia, Ascites, Gingival bleeding, Hemorrhoids, Loose stools, Tongue ulceration, Dysphagia, Oral soft tissue disorder NOS, Lip ulceration, Abdominal distension, Abdominal pain upper, Gastro-esophageal reflux disease, Glossodynia, Abdominal pain, Diarrhea, Oral pain, Toothache, Vomiting, gingival pain, upper gastrointestinal hemorrhage.

General disorders and administrative site disorders: Pyrexia, Edema peripheral, Rigors, Edema NOS, Pain NOS, Lethargy, Tenderness NOS, Fall, Chest discomfort, Intermittent pyrexia, Malaise, Crepitations NOS, Catheter site erythema, Catheter site pain, Injection site swelling, Asthenia, Chest pain, Chills, Fatigue, Mucosal inflammation, Edema, Pain, catheter site hemorrhage.

Hepatobiliary disorders: Hyperbilirubinemia, cholecystitis.

Infections and infestations: Pneumonia NOS, Cellulitis, Candidal infection NOS, Catheter related infection, Urinary tract infection NOS, Staphylococcal infection, Oral candidiasis, Sinusitis NOS, Bacteremia, Pneumonia, Sinusitis, Staphylococcal bacteremia, Tooth abscess, Upper respiratory tract infection, Urinary tract infection, fungal infection, sepsis, bronchopulmonary aspergillosis, peridiverticular abscess, respiratory tract infection, pseudomonal lung infection, Mycobacterium avium complex infection.

Injury, poisoning and procedural complications: Transfusion reaction, Abrasion NOS, Contusion, post procedural pain, post procedural hemorrhage.

Investigations: Cardiac murmur NOS, Blood alkaline phosphatase NOS increased, Aspartate aminotransferase increased, Blood urea increased, Blood lactate dehydrogenase increased, Blood albumin decreased, Blood bicarbonate increased, Blood chloride decreased, Protein total decreased, Blood bicarbonate decreased, Blood bilirubin decreased, Blood bilirubin increased, Breath sounds abnormal, Weight decreased

Metabolism and nutrition disorders: Hyperglycemia NOS, Hypoalbuminemia, Hypomagnesemia, Hypokalemia, Hyponatremia, Appetite decreased NOS, Anorexia, Hyperkalemia, Dehydration, Hyperglycemia.

Musculoskeletal and connective tissue disorders: Arthralgia, Pain in limb, Back pain, Chest wall pain. Musculoskeletal discomfort, Myalgia, Bone pain, Muscle spasms, Muscular weakness, musculoskeletal pain, Pain in extremity.

Nervous system disorders: Headache, Dizziness, Hypoesthesia, intracranial hemorrhage.

Psychiatric disorders: Insomnia, Confusional state, Anxiety, Depression, mental status changes.

Renal and urinary disorders: Dysuria, Urinary frequency, renal failure, urethral hemorrhage.

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Respiratory, thoracic and Mediastinal disorders: Cough, Pharyngitis, Crackles lung, Breath sounds decreased, Hypoxia, Rales, Postnasal drip, Dyspnea, Epistaxis, Pharyngolaryngeal pain, Pleural effusion, Sinus congestion, hemoptysis, lung infiltration, pulmonary embolism, respiratory arrest, pulmonary mass.

Skin and subcutaneous tissue disorders: Ecchymosis, Rash NOS, Erythema, Skin lesion NOS, Pruritus, Alopecia, Urticaria NOS, Swelling face, Dry skin, Night sweats, Petechiae, Rash, Skin lesion

Vascular disorders: Petechiae, Pallor, Hypotension NOS, Hematoma NOS, Hypertension, Hypotension

Allergic Reaction: Hypersensitivity (anaphylactic reaction).

No overall difference in safety was detected between patients > 65 years of age and younger patients in these myelodysplasia studies. No significant gender differences in safety or efficacy were detected.

Acute myeloid leukemia (AML)

Summary of the safety profile

The most common adverse drug reactions reported during treatment with decitabine are pyrexia, anaemia and thrombocytopaenia.

The most common Grade 3/4 adverse drug reactions included pneumonia, thrombocytopaenia, neutropaenia, febrile neutropaenia and anaemia.

It has been reported that patients treated with decitabine had adverse events with an outcome of death during treatment or within 30 days after the last dose of study drug.

In the decitabine treatment group, there was a higher incidence of treatment discontinuation due to adverse events in women compared to men.

Tabulated list of adverse drug reactions

Adverse drug reactions reported in AML patients treated with decitabine are summarised in Table below:

Within each frequency grouping, adverse drug reactions are presented in order of decreasing seriousness.

Table: Adverse drug reactions identified with Decitabine

System Organ Class	Frequency (all Grades)	Adverse Drug Reaction
Infections and infestations	Very common	pneumonia*
		urinary tract infection*
		All other infections (viral, bacterial, fungal)*, b, c, d
	Common	septic shock*

		sepsis ^a
		sinusitis
Blood and lymphatic disorders	Very common	febrile neutropaenia ^a
		neutropaenia ^a
		thrombocytopaenia ^{a, e}
		anaemia
		leukopaenia
	Uncommon	pancytopaenia ^a
Immune system disorders	Common	hypersensitivity including anaphylactic reaction ^f
Nervous system disorders	Very common	headache
Respiratory, thoracic and mediastinal disorders	Very common	epistaxis
	Not known	interstitial lung disease
Gastrointestinal disorders	Very common	diarrhoea
		vomiting
		nausea
	Common	stomatitis
	Not known	enterocolitis, including neutropaenic colitis, caecitis ^a
Skin and subcutaneous tissue disorders	Uncommon	acute febrile neutrophilic dermatosis (Sweet's syndrome)
General disorders and administration site conditions	Very common	pyrexia

^a Worst National Cancer Institute Common Terminology Criteria for Adverse Events Grade.

^b Excluding pneumonia, urinary tract infection, sepsis, septic shock and sinusitis.

^c The most frequently reported "other infections" were: oral herpes, oral candidiasis, pharyngitis, upper respiratory tract infection, cellulitis, bronchitis, nasopharyngitis.

^d Including enterocolitis infectious.

^e Including haemorrhage associated with thrombocytopaenia, including fatal cases.

^f Including preferred terms hypersensitivity, drug hypersensitivity, anaphylactic reaction, anaphylactic shock, anaphylactoid reaction, anaphylactoid shock.

* Includes events with a fatal outcome.

NA = Not applicable

Description of selected adverse drug reactions

Haematologic adverse drug reactions

The most commonly reported haematologic adverse drug reactions associated with decitabine treatment included febrile neutropaenia, thrombocytopaenia, neutropaenia, anaemia and leukopaenia.

Serious bleeding-related adverse drug reactions, some of which lead to a fatal outcome, such as central nervous system (CNS) haemorrhage and gastrointestinal (GI) haemorrhage, in the context of severe thrombocytopaenia, were reported in patients receiving decitabine.

Haematological adverse drug reactions should be managed by routine monitoring of complete blood counts and early administration of supportive treatments as required. Supportive treatments

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include, administration of prophylactic antibiotics and/or growth factor support (e.g., G-CSF) for neutropaenia and transfusions for anaemia or thrombocytopaenia according to institutional guidelines.

Infections and infestations adverse drug reactions

Serious infection-related adverse drug reactions, with potentially fatal outcome, such as septic shock, sepsis, pneumonia, and other infections (viral, bacterial and fungal) were reported in patients receiving decitabine.

Gastrointestinal disorders

Occurrences of enterocolitis, including neutropaenic colitis, caecitis have been reported during treatment with decitabine. Enterocolitis may lead to septic complications and may be associated with fatal outcome.

Respiratory, thoracic and mediastinal disorders

Cases of interstitial lung disease (including pulmonary infiltrates, organising pneumonia and pulmonary fibrosis) without signs of infectious aetiology have been reported in patients receiving decitabine.

Overdose

There is no direct experience of human overdose and no specific antidote. However, early clinical study data in published literature at doses greater than 20 times higher than the current therapeutic dose, reported increased myelosuppression including prolonged neutropaenia and thrombocytopaenia. Toxicity is likely to manifest as exacerbations of adverse drug reactions, primarily myelosuppression. Treatment for overdose should be supportive.

Pharmacological properties

Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agents, antimetabolites, pyrimidine analogues;

Mechanism of action

Decitabine (5-aza-2'-deoxycytidine) is a cytidine deoxynucleoside analogue that selectively inhibits DNA methyltransferases at low doses, resulting in gene promoter hypomethylation that can result in reactivation of tumour suppressor genes, induction of cellular differentiation or cellular senescence followed by programmed cell death.

Pharmacokinetic properties

The population pharmacokinetic (PK) parameters of decitabine were pooled from 3 clinical studies in 45 patients with AML or myelodysplastic syndrome (MDS) utilizing the 5-Day regimen. In each study, decitabine PK was evaluated on the fifth day of the first treatment cycle.

Distribution

The pharmacokinetics of decitabine following intravenous administration as a 1-hour infusion were described by a linear two-compartment model, characterised by rapid elimination of the drug from the central compartment and by relatively slow distribution from the peripheral compartment.

Decitabine exhibits linear PK and following the intravenous infusion, steady-state concentrations are reached within 0.5 hour. Based on model simulation, PK parameters were independent of time (i.e., did not change from cycle to cycle) and no accumulation was observed with this dosing regimen. Plasma protein binding of decitabine is negligible (< 1%). Decitabine $V_{d_{ss}}$ in cancer patients is large indicating distribution of the drug into peripheral tissues. There was no evidence of dependencies on age, creatinine clearance, total bilirubin, or disease.

Biotransformation

Intracellularly, decitabine is activated through sequential phosphorylation via phosphokinase activities to the corresponding triphosphate, which is then incorporated by the DNA polymerase. *In vitro* metabolism data and the human mass balance study results indicated that the cytochrome P450 system is not involved in the metabolism of decitabine. The primary route of metabolism is likely through deamination by cytidine deaminase in the liver, kidney, intestinal epithelium and blood. Results from the human mass-balance study showed that unchanged decitabine in plasma accounted for approximately 2.4% of total radioactivity in plasma. The major circulating metabolites are not believed to be pharmacologically active. The presence of these metabolites in urine together with the high total body clearance and low urinary excretion of unchanged drug in the urine (~4% of the dose) indicate that decitabine is appreciably metabolized *in vivo*. *In vitro* studies show that decitabine does not inhibit nor induce CYP 450 enzymes up to more than 20-fold of the therapeutic maximum observed plasma concentration (C_{max}). Thus; CYP-mediated metabolic drug interactions are not anticipated, and decitabine is unlikely to interact with agents metabolized through these pathways. In addition, *in vitro* data show that decitabine is a poor P-gp substrate.

Elimination

Mean plasma clearance following intravenous administration in cancer subjects was > 200 L/h with moderate inter-subject variability (Coefficient of variation [CV] is approximately 50%). Excretion of unchanged drug appears to play only a minor role in the elimination of decitabine.

Results from a mass balance study with radioactive ^{14}C -decitabine in cancer patients showed that 90% of the administered dose of decitabine (4% unchanged drug) is excreted in the urine.

Additional information on special populations

The effects of renal or hepatic impairment, gender, age or race on the pharmacokinetics of decitabine have not been formally studied. Information on special populations was derived from pharmacokinetic data from the 3 studies noted above, and from one Phase I study in MDS subjects, (N = 14; 15 mg/m² x 3-hours q8h x 3 days).

Older people

Population pharmacokinetic analysis showed that decitabine pharmacokinetics are not dependent on age (range studied 40 to 87 years; median 70 years).

Gender

Population pharmacokinetic analysis of decitabine did not show any clinically relevant difference between men and women.

Race

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Most of the patients studied were Caucasian. However, the population pharmacokinetic analysis of decitabine indicated that race had no apparent effect on the exposure to decitabine.

Hepatic impairment

The PK of decitabine have not been formally studied in patients with hepatic impairment. Results from a human mass-balance study and *in vitro* experiments mentioned above indicated that the CYP enzymes are unlikely to be involved in the metabolism of decitabine. In addition, the limited data from the population PK analysis indicated no significant PK parameter dependencies on total bilirubin concentration despite a wide range of total bilirubin levels. Thus, decitabine exposure is not likely to be affected in patients with impaired hepatic function.

Renal impairment

The PK of decitabine have not been formally studied in patients with renal insufficiency. The population PK analysis on the limited decitabine data indicated no significant PK parameter dependencies on normalized creatinine clearance, an indicator of renal function. Thus, decitabine exposure is not likely to be affected in patients with impaired renal function.

PHARMACEUTICAL INFORMATION

Incompatibilities

This medicinal product must not be mixed with other medicinal products except those mentioned in reconstitution procedure.

Storage: Store below 30°C

Keep all medicines out of reach of children

Dosage form: Powder for Solution for Injection

Presentation: One vial in a carton

Self-life

Unopened vials: 24 Months.

Protected from light.

Reconstituted and diluted solution

Within 15 minutes of reconstitution, the concentrate (in 10 ml of sterile water for injections) must be further diluted with cold (2°C - 8°C) infusion fluids. This prepared diluted solution for intravenous infusion can be stored at 2°C - 8°C for up to a maximum of 7 hours before administration.

From a microbiological point of view, the product should be used within the time period recommended above. It is the responsibility of the user to follow the recommended storage times and conditions and ensure that reconstitution has taken place in aseptic conditions.

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Manufactured by:

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M.L.No 41/VP/AP/2006/F/R

PRODUCT REGISTRATION HOLDER

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